6.3 Preserving Distance

Lesson Introduction

Benchmarks	MA.912.GR.2.2 Identify transformations that do or do not preserve distance. MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.	Learning Targets	• I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance. • I can draw a sequence of transformations to determine if the transformation does or does not preserve distance. • I can identify transformations that do or do not preserve distance.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.	N/A

Essential Questions	• How do you know if a sequence of transformations does or does not preserve distance? • What methods might you use to prove that the distance was preserved?
Lesson Narrative	This lesson explores sequences of transformation and whether the transformation preserved distance. Students will work with a partner to explore several quadrilaterals and their behavior under different transformations. They will “bring it together” through discovery of which transformation does or does not preserve distance.

Component Summary				
	6.3.1 Warm-Up (~5 min.)	Notice and wonder about a grid drawing *(SE p. 296)*	6.3.3 Bring It Together (5 min.)	Completing statements of transformations preserving distance *(SE p. 299)*
	6.3.2 Exploration (15-20 min.)	Discovering transformations that preserve distance *(SE p. 297)*	6.3.4 Guided Instruction (10-15 min.)	Determining if sides are congruent after performing transformations *(SE p. 299)*
	6.3.5 Wrap-Up (~5 min.)	Reflection using KWL *(SE p. 301)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Reflection • Rotation • Translation • Dilation • Congruent	• Patty Paper • (Optional) Graph Paper • (Optional) Digital Graphing Tool (such as Desmos or GeoGebra)	• Consider having the quadrilaterals from 6.3.2 Exploration question 3 graphed ahead of time if needed for the discussion.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may struggle recognizing which type of transformations occurred from a grid.
Students may not recognize the correct transformation if presented only the coordinates of the preimage and image.</td><td>Consider using Anchor Charts to illustrate the different types of transformations.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Take Turns
Before completing 6.3.3 Bring It Together, set students up to take turns in answering the questions. Set the expectation that each partner takes a statement and justifies their choices by referring to where it was discovered in 6.3.2 Exploration. This will build upon the expectation that good mathematicians seek answers in their texts and resources.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students will use patterns and structure to help them understand and connect the concept of rigid and nonrigid transformations. They will focus on the relevant details within the problem to determine which are and are not.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The use of patty paper in this lesson is a great visual and kinesthetic way to model preserving distance with transformations. You could also provide physical objects and spatial models to convey perspective or interaction with this lesson as well. Use of graphing technology could additionally be introduced to visually show how the transformation was accomplished.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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6.3.1 Warm-Up: Notice and Wonder

***Component Overview**: Students are asked to write down things they notice and wonder about the drawing of Mary Hawtrey.*

6.3.1 Warm-Up: Notice and Wonder

Artist John Hoskins sketched a drawing of Mary Hawtrey, also known as Lady Mary Bankes, in 1821. Henry Pierce Bone then used the sketch to create an enamel miniature painting.

Following the death of her husband, Lady Mary Bankes successfully defended the Corfe Castle in Dorset, England during a siege in 1643. Unfortunately, during a second siege in 1646 the castle was destroyed.

1. What do you notice about the sketch of Lady Mary Bankes?
   Answers will vary. Students may notice the signatures below the illustration, the background drawing drawn in proportion to invisible point in the distance (e.g. individual picture is bigger, landscape is smaller).

2. What do you wonder about the sketch of Lady Mary Bankes or her story?
   Answers will vary. I wonder what she is holding in her hand and whether it has any significance.

After the students have had the opportunity to write down things they notice and wonder, ask several students to share their observations. You could record these for all to see. Steer the conversation towards a mathematical discussion. This low-stake activity gets students thinking and responding, leading them into the context of the lesson, and might pique their curiosity.

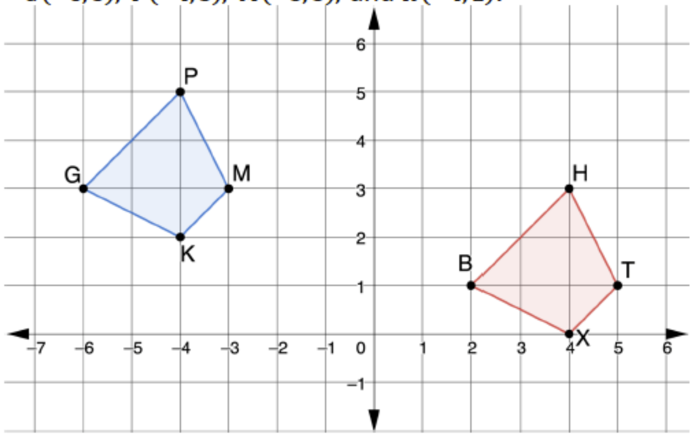
6.3.2 Exploration: Sequence of Transformations

Component Overview: *Students will work with a partner to determine what transformation(s) have happened in the questions. They will use patty paper to help connect the movement of the object and the transformation that was applied to it. They will then form a conjecture about the corresponding sides of the objects. Finally, they will confirm if distance was preserved after performing the sequence of transformations.*

6.3.2 Exploration: Sequence of Transformations

With your partner, answer the following questions.

1. $BHTX$ with vertices $B(2,1)$, $H(4,3)$, $T(5,1)$, and $X(4,0)$ is the result of a transformation of $GPMK$ with vertices $G(-6,3)$, $P(-4,5)$, $M(-3,3)$, and $K(-4,2)$.



a. What transformation occurred to $GPMK$ that resulted in $BHTX$?
$BHTX$ is a translation from the original image, it has been shifted 8 units to the right and 2 units down.

b. Use patty paper to trace $GPMK$, then place $GPMK$ onto $BHTX$.

c. What do you notice about the distances between the vertices of corresponding sides?
The distances between the vertices of $BHTX$ are the same as the distances between the vertices of the original quadrilateral $GPMK$.

Students may find the use of the patty paper helpful for question 1a and for 1b. Consider having the image copied so that it can be manipulated to determine the transformation. This will provide a kinesthetic entry-point into the question.

d. What can be concluded about the lengths of the corresponding sides after this transformation occurs?
The side lengths will remain constant if the transformation is a translation.

2. The coordinates of $RWTY$ and $R'W'T'Y'$ are shown in the table, where $RWTY$ is the preimage.

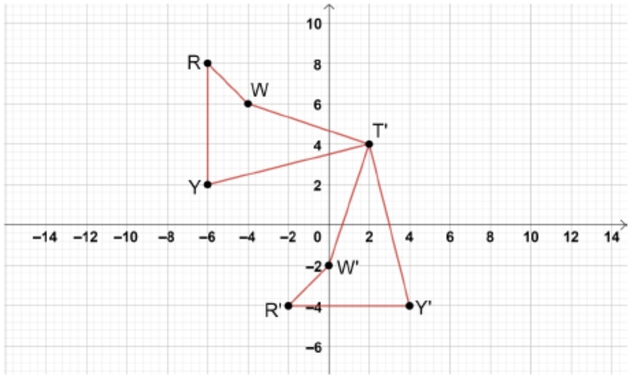
$RWTY$

Vertex	x	y
R	-6	8
W	-4	6
T	2	4
Y	-6	2

$R'W'T'Y'$

Vertex	x	y
R'	-2	-4
W'	0	-2
T'	2	4
Y'	4	-4

a. Use the coordinate grid to graph $RWTY$ and $R'W'T'Y'$.



b. What type of transformation was applied to the preimage, $RWTY$?
$R'W'T'Y'$ is a rotation of the preimage $RWTY$. The image has been rotated about point T $90°$ counterclockwise.

Collect and Display
Consider using the instructional routine Collect and Display after questions 1d, 2c, 3b, and 4b. Have students share their responses with the class, and display responses for all students to see. This will help students draw conclusions prior to 6.3.3 Bring It Together and help them make the connections about which transformations preserve distance.

6.3.2 Exploration: Sequence of Transformations (continued)

c. How could the coordinates of the preimage and image be used to show that the transformation preserved side length?
The coordinates can be used to find the distance of each side. When corresponding sides have the same distance, the transformation preserves side length.

3. The coordinates of $MPRD$ and $SGTW$ are shown in the table, where $MPRD$ is the preimage.

$MPRD$

Vertex	x	y
M	8	-7
P	9	-4
R	4	-2
D	1	-5

$SGTW$

Vertex	x	y
S	8	7
G	9	4
T	4	2
W	1	5

a. What transformation occurred from the preimage to the image?
There was a reflection across the x-axis. The x values have remained the same and the y values have been multiplied by -1.

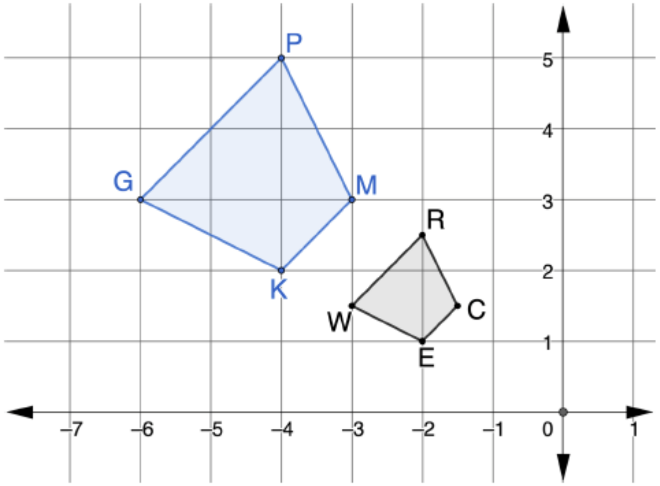
b. Explain how the coordinates of the preimage and image show that the transformation preserved distance.
Using the distance formula, the coordinates on $SGTW$ demonstrate that the side lengths are the same as on $MPRD$, this shows that the transformation preserved distance.

To encourage mathematical reasoning as students complete the transformations, prompt students with questions like:
- *What led you to the transformation you chose for this question?*
- *Could more than one transformation work in this question? Why or why not?*

Students may provide the answer to preserved distance using different words. Lead students to stating that the distance between the points remain the same. As this is a rotation, it would be less likely that students notice directly from the points.

Consider providing students with digital tools like GeoGebra or Desmos to complete question 3. This can provide kinesthetic and visual entry-points.

4. $WRCE$ is the result of a transformation on $GPMK$.



a. What transformation occurred to $GPMK$ that resulted in $WRCE$?
A dilation centered at the origin with a scale factor of $\frac{1}{2}$.

b. What can be concluded about the lengths of each corresponding side after this transformation occurs?
The lengths of corresponding sides are not the same.

Students who finish early could extend their thinking by creating a sequence of transformations of their own and exchanging with a partner to complete each other's problem.

6.3.3 Bring It Together: Preserving Distance

Component Overview: *Students will summarize the properties about transformations preserving distance by completing several statements.*

6.3.3 Bring It Together: Preserving Distance

1. Complete the statements based on the Exploration.

   a. $BHTX$ was created by [○ reflection. / ○ translation. / ○ rotation. / ○ dilation.] This transformation [○ does / ○ does not] preserve distance between the vertices.

   b. $R'W'T'Y'$ was created by [○ reflection. / ○ translation. / ○ rotation. / ○ dilation.] This transformation [○ does / ○ does not] preserve distance between the vertices.

   c. $SGTW$ was created by [○ reflection. / ○ translation. / ○ rotation. / ○ dilation.] This transformation [○ does / ○ does not] preserve distance between the vertices.

   d. $WRCE$ was created by [○ reflection. / ○ translation. / ○ rotation. / ○ dilation.] This transformation [○ does / ○ does not] preserve distance between the vertices.

Take Turns
Before completing these statements, set students up to take turns in answering the questions. Set the expectation that each partner takes a statement and justifies their choices by referring to where this was discovered in 6.3.2 Exploration. This will build upon the expectation that good mathematicians seek answers in texts and resources.

This page of the lesson is a phenomenal resource for students going forward with transformations and preserving distance. Consider having your students bookmark this page for quick access.

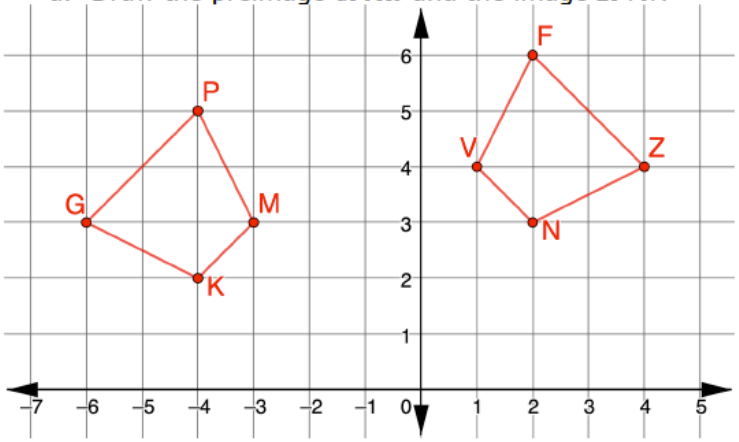
6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation

Component Overview: *Students will graph a preimage and image to determine what transformation has occurred and then justify whether the transformations preserve distance. Encourage students to use a variety of methods for determining congruence, such as patty paper and calculating the lengths of line segments.*

6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation

1. $GPMK$ with vertices $G(-6,3)$, $P(-4,5)$, $M(-3,3)$, and $K(-4,2)$ is reflected over $x=-2$, then translated right 2 units and up 1 unit to create $ZFVN$, respectively.

   a. Draw the preimage $GPMK$ and the image $ZFVN$.



   b. Use patty paper or the coordinates to determine if the corresponding sides of $GPMK$ and $ZFVN$ are congruent.
   The corresponding sides are congruent.

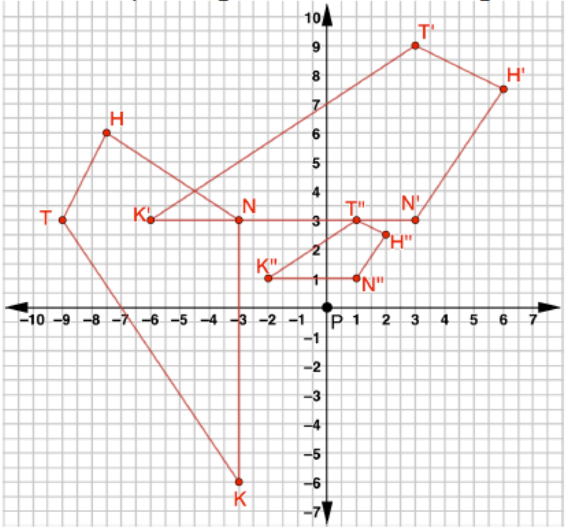
2. $THNK$ with vertices $T(-9,3)$, $H(-7.5,6)$, $N(-3,3)$, and $K(-3,-6)$ is rotated $90°$ clockwise about the origin, then dilated using the scale factor of $\frac{1}{3}$ centered at the origin to create $T''H''N''K''$, respectively.

Consider asking the following:

- *If you swapped the order of the transformations in question 1, would the result be the same image?*
- *What method did you use to determine if the corresponding sides were congruent? Why did you choose this method?*

6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation (continued)

a. Draw the preimage $THNK$ and the image $T''H''N''K''$.



b. Determine if the corresponding sides of $THNK$ and $T''H''N''K''$ are congruent.
No. The sides are not congruent.

c. Explain how you could use the description of a sequence of transformations to determine if the transformation does or does not preserve distance.
Answers will vary. You could use the description of the sequence to determine if distance would be preserved by using your knowledge of the types of transformations. The only transformation that changes side lengths is dilation.

d. How would your response differ if you were given the coordinates of a sequence of transformations?
Answers will vary. If I were given the coordinates of the transformations, I could use the distance between sides to determine if the distance was preserved. I also could use the points to draw the images and determine which transformations had occurred.

If students finish early, challenge them to determine if this sequence of transformations is commutative.

6.3.5 Wrap-Up: Reflection

Component Overview: *Students complete a KWL self-reflection on the lesson.*

6.3.5 Wrap-Up: Reflection

1. Complete the table.
   Answers will vary.

The terms and information used in the lesson that I knew were …	Something I wondered during the lesson was …	Something I learned during the lesson was …

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.